

KUCHING INTL, Malaysia

WMO: 964130

Lat: 1.485N

Lon: 110.347E

Elev: 27

StdP: 101.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	22.6	20.7	15.4	27.9	21.1	15.8	27.3	6.3	27.5	5.7	27.6	1.1	140	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.3	34.1	26.2	33.5	26.2	33.0	26.2	27.7	31.8	27.2	31.4	26.9	31.0	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	22.0	30.0	26.1	21.6	29.6	25.9	21.3	29.3	88.4	32.2	86.3	31.8	84.9	31.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.8	4.2	21.3	35.5	0.6	0.7	20.9	36.0	20.5	36.4	20.1	36.8	19.7	37.3		
(5)			20.6	28.7	0.6	1.0	20.1	29.5	19.8	30.0	19.4	30.6	19.0	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	26.6	26.6	27.3	27.8	28.2	28.0	27.9	27.9	27.7	27.5	27.4	27.0
	DBStd	1.11	1.08	0.99	0.96	0.85	0.95	1.03	1.09	1.14	1.09	0.94	0.79	0.94
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6383	514	466	536	536	564	539	554	554	530	542	521	528
	CDD18.3	3342	256	232	278	286	306	289	296	296	280	283	271	269
	CDH23.3	28789	1938	1858	2378	2514	2907	2705	2777	2772	2447	2278	2136	2079
(14) Wind	WSAvg	1.6	1.7	1.7	1.6	1.4	1.5	1.5	1.7	1.7	1.6	1.6	1.5	1.6
	PrecAvg	4208	687	505	350	288	254	229	199	238	258	343	355	488
	PrecMax	5290	1351	1559	748	565	432	542	446	547	451	623	610	937
	PrecMin	2771	146	119	149	84	87	100	48	22	95	143	157	145
	PrecStd	540	291	252	139	103	72	84	84	110	91	117	105	150
	DB	32.2	32.1	33.1	33.9	34.2	34.2	34.8	34.9	34.2	34.0	33.2	33.0	33.0
	MCWB	26.2	26.0	26.0	26.6	26.7	26.1	26.2	26.0	26.2	26.1	26.2	26.3	26.3
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	31.1	31.1	32.2	33.0	33.6	33.3	33.7	33.8	33.2	32.9	32.1	31.9	31.9
	2% MCWB	25.9	25.8	26.0	26.5	26.6	26.2	26.0	25.9	26.1	26.1	26.0	26.0	26.0
	5% DB	30.2	30.2	31.2	32.1	32.9	32.8	32.9	33.0	32.2	31.9	31.4	31.0	31.0
	5% MCWB	25.5	25.4	26.0	26.2	26.5	26.2	26.0	25.7	26.0	25.9	26.0	25.9	25.9
	10% DB	29.2	29.6	30.5	31.2	31.9	31.9	31.9	32.0	31.4	30.9	30.7	30.0	30.0
	10% MCWB	25.2	25.2	25.8	26.2	26.4	26.1	25.7	25.7	25.9	25.9	25.9	25.6	25.6
	WB	26.9	26.8	27.4	27.7	28.4	28.0	27.7	27.2	27.9	27.2	27.3	27.2	27.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	30.6	30.3	31.0	32.0	32.6	32.4	32.3	31.8	31.9	31.6	31.2	30.8	30.8
	2% WB	26.3	26.2	26.7	27.1	27.6	27.2	27.0	26.7	27.0	26.7	26.6	26.6	26.6
	2% MCDB	29.5	29.4	30.4	31.2	31.7	31.4	31.5	31.3	31.3	30.8	30.5	30.2	30.2
	5% WB	26.0	25.9	26.2	26.7	27.0	26.7	26.5	26.4	26.5	26.3	26.2	26.2	26.2
	5% MCDB	28.9	28.9	29.7	30.5	31.1	30.7	30.8	30.8	30.7	30.3	30.0	29.6	29.6
	10% WB	25.6	25.6	26.0	26.3	26.6	26.3	26.1	26.1	26.1	26.0	26.0	25.8	25.8
	10% MCDB	28.3	28.4	29.2	29.9	30.4	30.2	30.3	30.4	30.1	29.8	29.6	29.0	29.0
(35) Mean Daily Temperature Range	MDBR	6.2	6.3	7.2	8.1	8.3	8.4	8.6	8.7	8.3	8.0	7.8	7.0	7.0
	5% DB	7.5	7.5	8.4	9.0	9.4	9.5	9.7	9.9	9.5	9.2	8.6	8.1	8.1
	5% MCWBR	2.9	2.9	3.2	3.4	3.5	3.5	3.5	3.6	3.7	3.6	3.4	3.2	3.2
	5% WB	6.8	6.7	7.7	8.3	8.6	8.7	8.9	9.1	8.8	8.4	8.0	7.5	7.5
	5% MCWBR	2.9	2.9	3.2	3.5	3.6	3.6	3.6	3.6	3.6	3.8	3.5	3.4	3.2
(40) Clear-Sky Solar Irradiance	taub	0.422	0.430	0.430	0.415	0.420	0.418	0.438	0.527	0.565	0.549	0.427	0.413	0.413
	taud	2.450	2.406	2.398	2.441	2.428	2.443	2.381	2.111	2.004	2.040	2.419	2.478	2.478
	Ebn at Noon	906	902	896	891	864	853	841	784	771	789	895	909	909
	Edh at Noon	117	124	125	117	115	111	119	161	183	177	120	113	113
(44) All-Sky Solar Radiation	RadAvg	4.11	4.40	4.94	5.08	4.96	4.83	4.94	4.92	4.54	4.45	4.53	4.30	4.30
	RadStd	0.40	0.30	0.26	0.20	0.22	0.20	0.29	0.26	0.44	0.56	0.23	0.28	0.28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.37	+0.23	+0.22	+0.55	+0.59	+0.64	N/A	N/A	+153	+153
(47) Regional (1 neighbor)		+0.37	+0.23	+0.22	+0.55	+0.59	+0.64	N/A	N/A	+153	+153

Nomenclature: See separate page